

Norway

SLEIPNER

Volve F

F-7

F-7

Design: F-7

Standard Survey Report

11 april, 2018

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	Rotary Table @ 54.90m (Rotary Table (As Installed))
Site:	Volve F	MD Reference:	Rotary Table @ 54.90m (Rotary Table (As Installed))
Well:	F-7	North Reference:	Grid
Wellbore:	F-7	Survey Calculation Method:	Minimum Curvature
Design:	F-7	Database:	Production EDM P246N

Project	SLEIPNER, Norway		
Map System:	Universal Transverse Mercator	System Datum:	Mean Sea Level
Geo Datum:	European 1950 - Mean		Using Well Reference Point
Map Zone:	Zone 31N (0 E to 6 E)		Using geodetic scale factor

Site	Volve F, 15/9			
Site Position:		Northing:	6,478,563.53 m	Latitude: 58° 26' 29.8068 N
From:	Map	Easting:	435,050.02 m	Longitude: 1° 53' 14.9285 E
Position Uncertainty:	0.00 m	Slot Radius:	13.200 in	Grid Convergence: -0.95 °

Well	F-7			
Well Position	+N/-S	1.96 m	Northing:	6,478,565.48 m
	+E/-W	-1.11 m	Easting:	435,048.91 m
Position Uncertainty	0.00 m	Wellhead Depth:	91.00 m	Water Depth: 91.00 m

Wellbore	F-7				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	NETICREFERENCE	08.11.2005	-2.54	71.62	50,283

Design	F-7				
Audit Notes:	Bit TD changed from 1083 mMD to 1085 mMD according to DBR.				
Version:	1.0	Phase:	ACTUAL	Tie On Depth:	145.90
Vertical Section:	Depth From (TVD) (m)	+N/-S (m)	+E/-W (m)	Direction (°)	
	145.90	1.96	-1.11	248.17	

Survey Program	Date	06.08.2012			
From (m)	To (m)	Survey (Wellbore)	Tool Name	Description	
160.00	317.00	F7 Single Shot Surveys (F-7)	Wellbore Surveyor, stat	Gyro Tool from GD	
349.63	890.69	F7 GWD Surveys (F-7)	Wellbore Surveyor, stat	Gyro Tool from GD	
962.41	1,083.00	12 1/4" Surveys (F-7)	Magnetic, std, non-mag	Magnetic Tools (MWD, EMS)	

Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
145.90	0.00	0.00	145.90	1.96	-1.11	0.00	0.000	0.000	0.000	
160.00	0.23	274.41	160.00	1.96	-1.14	0.03	0.489	0.489	0.000	
170.00	0.26	272.78	170.00	1.96	-1.19	0.06	0.092	0.090	-4.890	
180.00	0.30	262.14	180.00	1.96	-1.23	0.11	0.196	0.120	-31.920	
190.00	0.22	267.67	190.00	1.95	-1.28	0.15	0.251	-0.240	16.590	
199.00	0.32	260.35	199.00	1.95	-1.32	0.19	0.352	0.333	-24.400	
208.00	0.31	226.78	208.00	1.93	-1.36	0.24	0.607	-0.033	-111.900	

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Survey									
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
218.00	0.40	244.76	218.00	1.89	-1.41	0.30	0.426	0.270	53.940
229.00	0.25	249.93	229.00	1.87	-1.47	0.36	0.416	-0.409	14.100
240.00	0.46	229.35	240.00	1.83	-1.53	0.43	0.661	0.573	-56.127
250.00	0.24	304.09	250.00	1.82	-1.58	0.48	1.378	-0.660	224.220
259.00	0.22	334.00	259.00	1.84	-1.60	0.49	0.401	-0.067	99.700
270.00	0.06	63.36	270.00	1.87	-1.60	0.49	0.620	-0.436	243.709
280.00	0.31	14.41	280.00	1.89	-1.59	0.47	0.823	0.750	-146.850
289.00	0.33	42.81	289.00	1.94	-1.57	0.43	0.527	0.067	94.667
296.00	0.22	272.12	296.00	1.95	-1.57	0.42	2.151	-0.471	-560.100
305.00	0.26	296.43	305.00	1.96	-1.60	0.45	0.361	0.133	81.033
317.00	0.24	247.57	317.00	1.96	-1.65	0.49	0.519	-0.050	-122.150
349.63	0.44	280.87	349.63	1.96	-1.84	0.67	0.251	0.184	30.616
359.37	0.42	300.40	359.37	1.99	-1.90	0.72	0.453	-0.062	60.154
368.26	0.39	302.91	368.26	2.02	-1.96	0.76	0.118	-0.101	8.470
387.94	0.60	271.39	387.94	2.06	-2.12	0.89	0.513	0.320	-48.049
408.58	0.40	289.90	408.58	2.09	-2.29	1.05	0.370	-0.291	26.904
437.44	0.63	262.49	437.43	2.10	-2.55	1.27	0.344	0.239	-28.493
465.09	0.58	263.80	465.08	2.06	-2.83	1.56	0.056	-0.054	1.421
489.12	2.21	273.05	489.11	2.08	-3.42	2.09	2.048	2.035	11.548
516.00	2.54	280.07	515.96	2.21	-4.52	3.07	0.490	0.368	7.835
545.92	2.89	264.68	545.85	2.25	-5.93	4.36	0.807	0.351	-15.431
569.26	3.78	268.37	569.15	2.18	-7.28	5.64	1.176	1.144	4.743
609.63	5.24	263.11	609.39	1.92	-10.44	8.67	1.127	1.085	-3.909
650.56	7.38	259.43	650.07	1.21	-14.88	13.06	1.596	1.569	-2.697
690.34	9.04	261.55	689.44	0.28	-20.49	18.60	1.272	1.252	1.599
730.15	8.46	260.24	728.79	-0.67	-26.46	24.51	0.462	-0.437	-0.987
770.78	9.53	256.89	768.92	-1.94	-32.69	30.76	0.879	0.790	-2.474
809.78	9.32	258.24	807.39	-3.32	-38.92	37.06	0.235	-0.162	1.038
851.55	9.22	259.74	848.62	-4.60	-45.53	43.67	0.188	-0.072	1.077
890.69	8.60	264.09	887.29	-5.46	-51.52	49.55	0.701	-0.475	3.334
962.41	8.30	261.78	958.23	-6.75	-61.98	59.74	0.189	-0.125	-0.966
1,005.41	8.53	262.86	1,000.77	-7.59	-68.22	65.84	0.195	0.160	0.753
1,043.96	8.66	263.46	1,038.88	-8.28	-73.94	71.41	0.123	0.101	0.467
1,083.00	8.66	263.46	1,077.48	-8.95	-79.78	77.08	0.000	0.000	0.000

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Site:	Volve F	MD Reference:	Rotary Table @ 54.90m (Rotary Table (As Installed))
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Design Targets									
Target Name	Dip Angle	Dip Dir.	TVD	+N/-S	+E/-W	Northing	Easting	Latitude	Longitude
- hit/miss target	(°)	(°)	(m)	(m)	(m)	(m)	(m)		
- Shape									
Utsira Top F7 071106	0.00	0.00	882.90	-7.53	-50.04	6,478,556.00	435,000.00	58° 26' 29.5369 N	1° 53' 11.8524 E
- actual wellpath misses target center by 2.29m at 886.17m MD (882.82 TVD, -5.39 N, -50.85 E)									
- Point									
Utsira Top F7 v2 071106	0.00	0.00	884.90	6.48	-100.06	6,478,570.00	434,950.00	58° 26' 29.9627 N	1° 53' 8.7555 E
- actual wellpath misses target center by 49.82m at 895.19m MD (891.74 TVD, -5.53 N, -52.19 E)									
- Point									
Utsira Base	0.00	0.00	1,065.90	15.38	-417.67	6,478,578.90	434,632.50	58° 26' 30.0798 N	1° 52' 49.1724 E
- actual wellpath misses target center by 338.96m at 1083.00m MD (1077.48 TVD, -8.95 N, -79.78 E)									
- Point									

Casing Points					
Measured Depth	Vertical Depth			Casing Diameter	Hole Diameter
(m)	(m)	Name		(in)	(in)
202.00	202.00	30"		30.000	36.000
907.70	904.11	13 3/8"		13.375	17.500
1,062.00	1,056.72	9 5/8"		9.625	12.250

Design Annotations					
Measured Depth	Vertical Depth	Local Coordinates			
(m)	(m)	+N/-S	+E/-W	Comment	
(m)	(m)	(m)	(m)		
1,083.00	1,077.48	-8.95	-79.78	Projection	

Checked By: _____ Approved By: _____ Date: _____